Project Title

Develop a String Matching Application Using the Z-Algorithm for Efficient Pattern Searching

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

This project presents a string matching application based on the Z-Algorithm for efficient pattern searching in large text datasets. The application preprocesses a combined string consisting of the pattern, a delimiter, and the text to construct the Z-array. By analyzing the Z-array, all occurrences of the given pattern in the text can be identified efficiently. The project also implements the Naïve String Matching Algorithm to compare the performance of both techniques. The comparison focuses on execution time and efficiency for different text and pattern sizes. While the Naïve approach may perform repeated character comparisons and has a worst-case time complexity of $O(n \times m)$, the Z-Algorithm performs pattern matching in linear time with a complexity of $O(n + m)$, where n is the text length and m is the pattern length. The project demonstrates the practical advantages of the Z-Algorithm for fast, scalable, and efficient string searching.

Signature of the Guide